

The Social Dynamics of Voice Cloning: Trust, Privacy, and Ethical Tensions When Sharing Your AI Voice Replica

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Abstract

Rapid voice cloning enables the creation and sharing of Human Voice Cloning Agents (HVCAs), pushing this technology into public use and raising complex social and ethical issues alongside significant attention. This study investigates user experiences, perceptions, and impacts during HVCA creation, sharing, and interaction. Employing interviews, survey, and social media analysis, the research provides multi-dimensional empirical insights. Findings indicate HVCA sharing creates complex trust dynamics (familiarity vs. risk concerns), blurs human-AI boundaries affecting relationships, and poses challenges in voice identity control, privacy, and ethical responsibility. This research offers initial empirical insights into HVCAs in user-driven social contexts, highlighting unique trust, privacy, and ethical challenges. The findings have crucial implications for responsible HVCA design, platform policies, and regulation, emphasizing enhanced control, privacy, ethics, and oversight.

CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; *Empirical studies in collaborative and social computing*; • **Security and privacy** → Social aspects of security and privacy.

Keywords

voice cloning; ethics; responsible AI; conversational agents

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1 Introduction

Historically, self-voice cloning required extensive audio input. Now, advancements in neural networks allow voice cloning with only seconds of audio, enabling fine control over speech characteristics at low cost [24]. Developers are integrating this rapid voice cloning with Large Language Model (LLM) dialogue agents to create Human Voice Cloning Agents (HVCAs), allowing users to build and share AI agents with their own voice (Figure 1a, 1b). This ease of creation and sharing is accelerating HVCA adoption, moving the technology into daily social scenarios, as seen in viral social media content. HVCAs introduce unique complexities beyond traditional AI tools, potentially acting as implicit identities participating in social interactions.

The creation, sharing, and usage of HVCAs pose complex challenges for creators and listeners, including understanding AI trust with voice identity, defining privacy boundaries, and navigating ethical dilemmas of control and responsibility. While existing AI ethics research addresses identity replication, social participation, misuse, relationship changes, privacy, and trust [12, 20, 28], there's a lack of specific, in-depth research on HVCAs, which combine personalized voice identity, advanced dialogue capabilities, and active social sharing. Specifically, user experiences during HVCA creation/sharing and public reactions to their social use are not well understood.

Addressing this gap, this study explores two research questions: RQ1: User experiences and perceptions during HVCA creation/sharing regarding identity, trust, and ethics. RQ2: How HVCA sharing in social interaction affects trust dynamics, interpersonal relationships, and public opinion.

2 Background and Related Work

Conversational AI platforms in China (e.g., Cici AI [3], Kimi [4], Qwen [5]) and popular AI voice cloners elsewhere allow users to quickly clone their voices with minimal audio. This functionality is even integrated into smart car voice assistants. Combining these

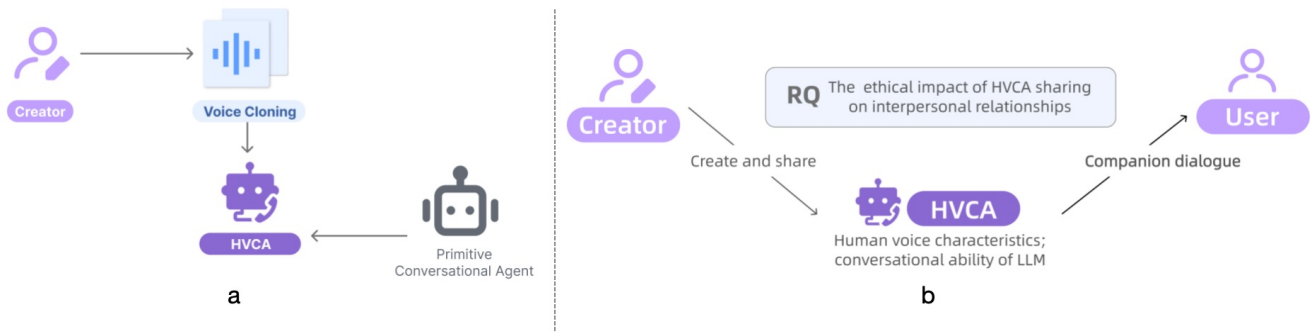


Figure 1: Schematic diagram of HVCA creation and sharing.

cloned voices with LLMs creates Human Voice Cloning Agents (HVCAs) capable of fluent conversation. Mobile AI software includes sharing mechanisms, promoting HVCA use within social networks and raising complex issues of identity, trust, privacy, and ethics. By simulating real voices and interacting human-like, HVCAs "represent" individuals digitally.

While related concepts exist (Deepfake, Digital Thought Clone), HVCAs are distinct. We define an HVCA as an AI agent actively created by a user using their personal voice samples. Leveraging few-shot learning neural networks, these systems can generate speech with the user's vocal identity from minimal audio input[7][15]. This voice synthesis is then combined with LLMs for interactive conversation, allowing the agent to not only sound authentic but also engage fluently on a wide range of topics. Unique characteristics positioning HVCAs in social computing and HCI include being based on a real individual's voice identity, user-driven creation/dissemination, and application in social companionship/daily communication. These traits necessitate deep exploration of the issues they raise[6].

Psychological research shows familiar voices enhance recognition, understanding, memory, and emotional resonance [13, 18]. In HCI, familiar voices increase anthropomorphism and trust [26][19], though this can lead to over-reliance [23]. When users create and share HVCAs, the familiar voice may boost identification with the agent [21][16]. AI-involved social interaction is a growing research area, exploring issues like lack of social translucence and privacy from AI accessing digital footprints [28], and how AI tools like smart replies can negatively impact interpersonal relationships despite convenience [12].

This complexity can be understood through the lens of identity elasticity, which distinguishes between stretching a self-narrative to maintain continuity and creating a break with the past self [10][25]. HVCAs uniquely embody this tension: the creator's voice provides narrative continuity, while the independent AI-driven intellect represents a clear identity break. This paradox helps frame the complex dynamics of trust, boundaries, and control that emerge when HVCAs are shared. When a shared HVCA is used, trust dynamics become complex[8]. Listeners might over-trust the AI due to voice familiarity, blurring boundaries, while the creator fears misuse. Privacy concerns with HVCA sharing involve multiple parties,

and including personal identity in LLM dialogue challenges traditional boundaries. HVCAs offer a "safe space" for interaction but introduce dilemmas of identity borrowing and potential shifts in expectations between the user and creator regarding the human-AI relationship [20]. While AI trust, privacy, control, and social impact are studied, the interconnected, dynamic nature of these issues within the real user experience and public perception of user-created, voice-identified, socially-shared HVCAs remains a gap needing systematic investigation.

3 Methodology

This section details the mixed-methods approach used to investigate the perceptions of HVCA creators and the public, covering data collection and analysis, and noting IRB approval.

Data Collection Data was collected in two parts: An online questionnaire was distributed across Chinese social media platforms to understand HVCA usage and views. It included demographic questions, 5-point Likert scale items assessing trust and perceived risk, and open-ended questions about sharing practices. Of the 164 completions, 122 participants provided detailed views on HVCAs. To analyze public discourse, we collected 797 comments from Bilibili[2]. We used keyword searches (e.g., 'AI voice clone,' 'replicate voice') to identify four popular videos demonstrating HVCA use, and then extracted the comments for thematic analysis.

Semi-structured Interview Interview participants were recruited from the 22 survey respondents who indicated they had cloned their voice and shared their HVCA. Six individuals (3 male, 3 female, aged 22-40) agreed to be interviewed. These participants, who reported varying levels of technical savviness, used AI software like Cici AI [3], Kimi [4], Qwen [5], and AITO[1] car voice assistant. They primarily shared their HVCAs with close relations (parents, children, friends, partners). Semi-structured interviews followed a guide, allowing for in-depth discussion, with full audio recordings made with consent.

Data Analysis Reliability-coded thematic analysis was used for interview data, ensuring reliability and repeatability. Three independent researchers familiarized themselves with transcripts, survey responses, and social media comments. Preliminary coding identified segments related to research questions, and inconsistencies were resolved through cross-validation and discussion, refining the codebook. Iterative analysis integrated coding results to form

broader themes and sub-themes, which were clearly defined. Thematic analysis results from all three sources (with interviews as primary) were cross-referenced and compared to form comprehensive findings, more robustly answering the research questions.

4 Findings

Through the thematic analysis by two researchers, three themes in sharing with the HVCA were identified: trust, anticipatory change, and control[9].

4.1 Trust Dynamics from the Perspective of Creators, Users, and the Public

Trust is a complex core issue for Human Voice Cloning Agents (HVCAs) across creation, sharing, usage, and public discourse. This study found contradictory attitudes towards HVCA trust in both individual user experiences and public perception. Creators exhibited this contradiction when switching between voice creator and listener roles and across different stages of engagement. Some interview participants (T2-T4, T6) expressed heightened concern about voice privacy upon realizing cloning capability, fearing malicious use or fraud, and felt uneasy sharing their voice[17].

“Once I realized that it can clone my voice, I started to worry about my privacy. Will someone do something bad with my voice or even my face information? Since the device’s camera also recognizes my face, my concerns grew.” (P3)

“I am more afraid that my mom and dad will be cheated and if someone calls them with my voice in the future, it will be dangerous. Because my mom is very timid, she can be tricked easily.” (P6)

“If the voice is for my own use, I will trust it; but if it is borrowed by someone else, it doesn’t feel good.” (P2)

This concern about privacy and risks is widespread, echoed in online surveys and social media comments. Key trust themes were potential risks, regulation, and impact on relationships, with comments noting how such technology lowers the cost of illegal acts and could cause a crisis of trust. Some participants found their own voice made the AI more familiar, increasing their likelihood to trust it. Sharing the voice-carrying agent with familiar individuals highlighted that voice identity makes listeners more trusting, manifesting in two ways: 1) Listeners might mistake the HVCA for the real person. 2) Due to the AI’s appropriation of the real voice, users were more likely to trust the HVCA’s spoken content.

“When I’m not at home and my grandmother is lonely, she talks to the Agent with my voice all the time.” (P1)

However, this “closeness” didn’t always yield positive trust.

“Children may be very dependent on it because the attachment to mom is innate.” (P4)

The confusion of HVCAs with real people and the transference of feelings suggest using a real voice identity makes listeners more likely to trust the HVCA-person mapping and the HVCA’s content. Surveys and social media comments echoed these concerns, noting potential impacts on communication and fears of recording during other speech, leading to identity replication issues.

4.2 Impact of Sharing HVCA on Social Relationships and Interpersonal Expectations

Sharing and using HVCAs significantly impacts listeners’ expectations of the voice owner and their social relationship, evident in user experiences and public discussion. Initial sharing motivation was often curiosity, aiming for “intimate” companionship. Interviews showed diverse contexts, recipients, and uses among participants (T1-T6), but the process was consistently: clone voice, experience dialogue, feel surprise, have familiar people check similarity, and let them try cloning. Participants felt they unconsciously “deployed” the HVCA to familiar people and later became curious about the listener’s conversations with the HVCA[11].

“When spending time together in person, one would communicate with family members frequently enough not to be curious about what they said; when there was physical distance, one would be curious.” (P1)

“Very curious and will interpret the conversation as something the listener is trying to say to him/herself.” (P3)

“Will be curious, but does not want to be in full control because the listener has his or her own privacy.” (P5)

Due to the cloned voice, participants associated listener-HVCA conversation content with themselves. As P3 noted, they perceived it as the listener conveying something they wanted to say to the creator but did so through the HVCA due to companionship limitations. T4 and T6 were less curious about listener conversations, anticipating topics, and felt frequent real-world communication reduced curiosity. Social media comments showed widespread curiosity and desire to try cloning/using HVCAs with friends. Interview analysis revealed users recognized the HVCA’s LLM capabilities (patience, knowledge, emotional expression), potentially altering their expectations of the creator.

“HVCA has a better way of speaking and will say novel content.” (P1)

“It is more patient than a human and does not get tired of whatever is asked. Children may feel that the HVCA behaves better than a real parent when it comes to storytelling.” (P4)

Some participants (P3, P5) were both HVCA sharers and listeners. P3 preferred confiding in a friend’s HVCA over the friend directly, finding the format similar and blurring the line between the AI and the real person. Disappointment with the HVCA negatively affected his relationship with the friend, as he treated the HVCA as the friend, reduced direct chats, and became frustrated with the real friend. This blurring and transfer of emotions into real relationships was explicitly noted by P3 and strongly supported in public discussions about “blurred boundaries” and an “authenticity crisis.” While views on the impact of HVCAs on human-AI relationships varied, both users and the public agreed they would influence these relationships.

4.3 Control and Ethical Challenges in HVCA Sharing

Sharing HVCA also triggers user struggles for control over AI possessing their voice identity and complex ethical considerations, present in individual experiences and public attention. Interview findings showed creators' strong, complex desire for control over their HVCA.

"Sometimes I want to control the content of its speech, and I would like it to at least not answer certain unfriendly questions so that I would feel more comfortable sharing them with others. As the owner of the voice, I should have more control over it, and I should be able to decide whether the content of its conversation with the listener, can be saved or not." (P1)

Participants intuitively desired control over HVCA responses, fearing unfriendly answers could damage their voice identity. They also felt they should control chat log saving to protect this identity, considering others involved. This desire stems from perceiving voice as a personal identity marker and property, a view echoed in public discussions questioning voice property rights and the problems of misuse. Rooted in recognizing voice value, users' control demand aims to prevent risks like illegal use or impersonation fraud, concerns shared by the public. Creators differentiated control and disclosure based on sharing scope (familiar vs. public), believing informed consent works with familiar people based on trust[14].

"I will not share my HVCA on a public platform, and I should not know the content of strangers' conversations myself. In that case, the platform should not allow us ordinary people to make our HVCA public. But family members can use it, and I would like to know what they say, so I should make it clear to my family members at the beginning of the use that their conversations may be seen by me." (P4)

"I can have no control over the HVCA, but I would like to have some contents of their conversations shared with me, such as health status, etc., of course this is with the authorization of the other party." (P5)

Challenges regarding HVCA voice identity owners' control, right to know, and concerns cannot be solved by simple informed consent. P5's wish to balance respecting listener privacy with obtaining key information illustrates a boundary difficulty. This attempt to define control/knowledge based on sharing scope highlights privacy boundary challenges and control complexity introduced by HVCA sharing. Public comments on "blurred boundaries" indirectly reflect the difficulty in exercising control and protecting privacy without clear lines[22]. The control dilemmas and ethical considerations faced by creators are mirrored in broader public opinion under the theme of ethical/social fairness. Public comments raised technical ethics issues like voice authenticity and consent and recognized the difficulty in assigning responsibility for HVCA misuse, leading to widespread calls for norms and regulations.

5 Discussion

Our research reveals that HVCA, by combining a user's personal voice identity with AI attributes, create a fundamental social paradox. The familiar voice fosters intimacy and trust, yet the AI's autonomy simultaneously raises deep concerns about misuse, privacy, and authenticity, leading to what the public terms "boundary blurring" and an "authenticity crisis." This core tension—balancing the benefits of a novel social connection against its inherent risks—disrupts traditional models of trust and control, prompting calls for solutions that move beyond simple informed consent to encompass technological, regulatory, and educational frameworks.

These findings inform key implications for HVCA development, all stemming from the technology's tension between providing narrative continuity through voice and creating an identity break via its AI intellect. Design should enhance user control over this 'elastic' voice identity; platforms must establish clear responsibilities for this dual human-AI entity; and social education should be strengthened[27]. Our study's generalizability is limited by a small interview sample (N=6) from Chinese platforms and social media analysis confined to Bilibili[2], making our findings culturally situated. Future work should therefore involve larger-scale, cross-cultural surveys and longitudinal studies. For instance, one could investigate how these agents function across different scenarios, from public-facing roles to private tools for exploring self-identity.

6 Conclusion and Future Work

Through a combined analysis of interviews, questionnaires, and social media comments, this study reveals the complex challenges arising from users creating and sharing Human Voice Cloning Agents (HVCA): the two-sided nature of trust (familiarity attraction vs. risk concern), the impact on social relationships and expectations (boundary blurring and relationship changes), and dilemmas regarding control and ethics (difficulty in individual control and lack of norms). The study offers some of the first empirical insights into user experience and public perception in HVCA sharing scenarios, identifying challenges surrounding voice identity, sharing, and multi-party interests. These findings provide implications for HVCA design, platform policies, and regulations, emphasizing the necessity of enhanced control, privacy protection, ethical norms, and regulation. Despite limitations in sample and scope, this study lays a foundation for understanding the interplay of emerging AI, human interaction, and ethics, and indicates directions for future research.

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